

Week 1: Introduction to Prompt Engineering

Lecturer: Dr. Mehmet Ali Akyol

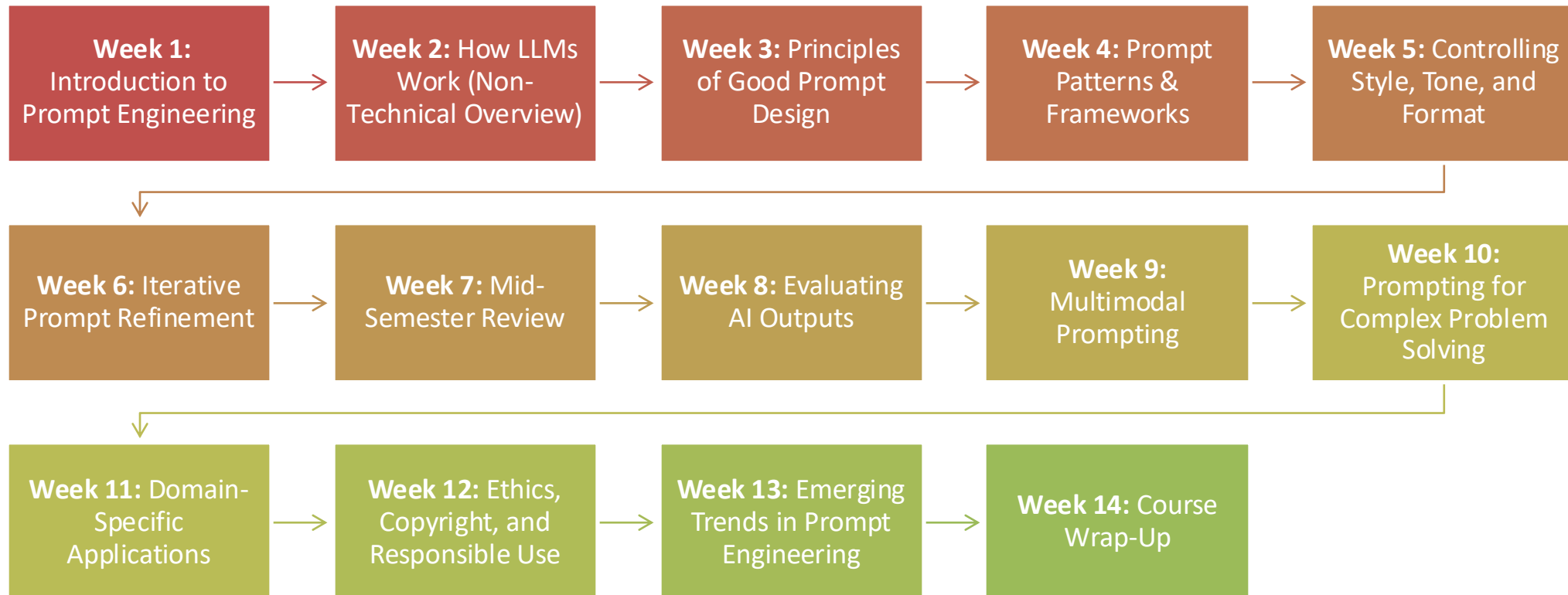
Course: Prompt Engineering



Course Overview

-   Course: Prompt Engineering
 -  Aim: Equip students with skills to design, refine, and evaluate prompts
 -  Open to all departments and disciplines
- 

Syllabus at a Glance



Learning Outcomes



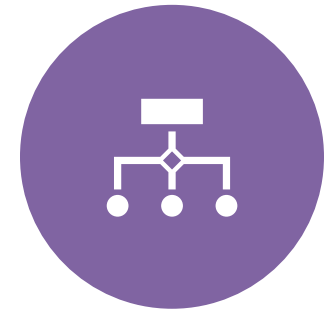
✦ CREATE CLEAR,
CONTEXT-AWARE PROMPTS



✦ USE STRUCTURED
PROMPTING STRATEGIES



✦ ASSESS AI RESPONSES
(ACCURACY, BIAS, TONE,
RELEVANCE)



✦ APPLY TECHNIQUES
ACROSS TEXT, IMAGE,
CODE, AND DATA

Grading & Evaluation



Weekly Quizzes: 30%



Midterm Exam: 30%



Final Exam: 40%

A background image showing a group of people with their hands raised in a gesture of introduction or agreement. The image is dark and has a white border. The text "Introduce yourselves!" is centered in the image.

Introduce yourselves!

Learning Objectives

1



Define what a prompt is

2



Understand its role across fields

3



Learn key terms: Prompt, LLM, Generative AI, Context

4

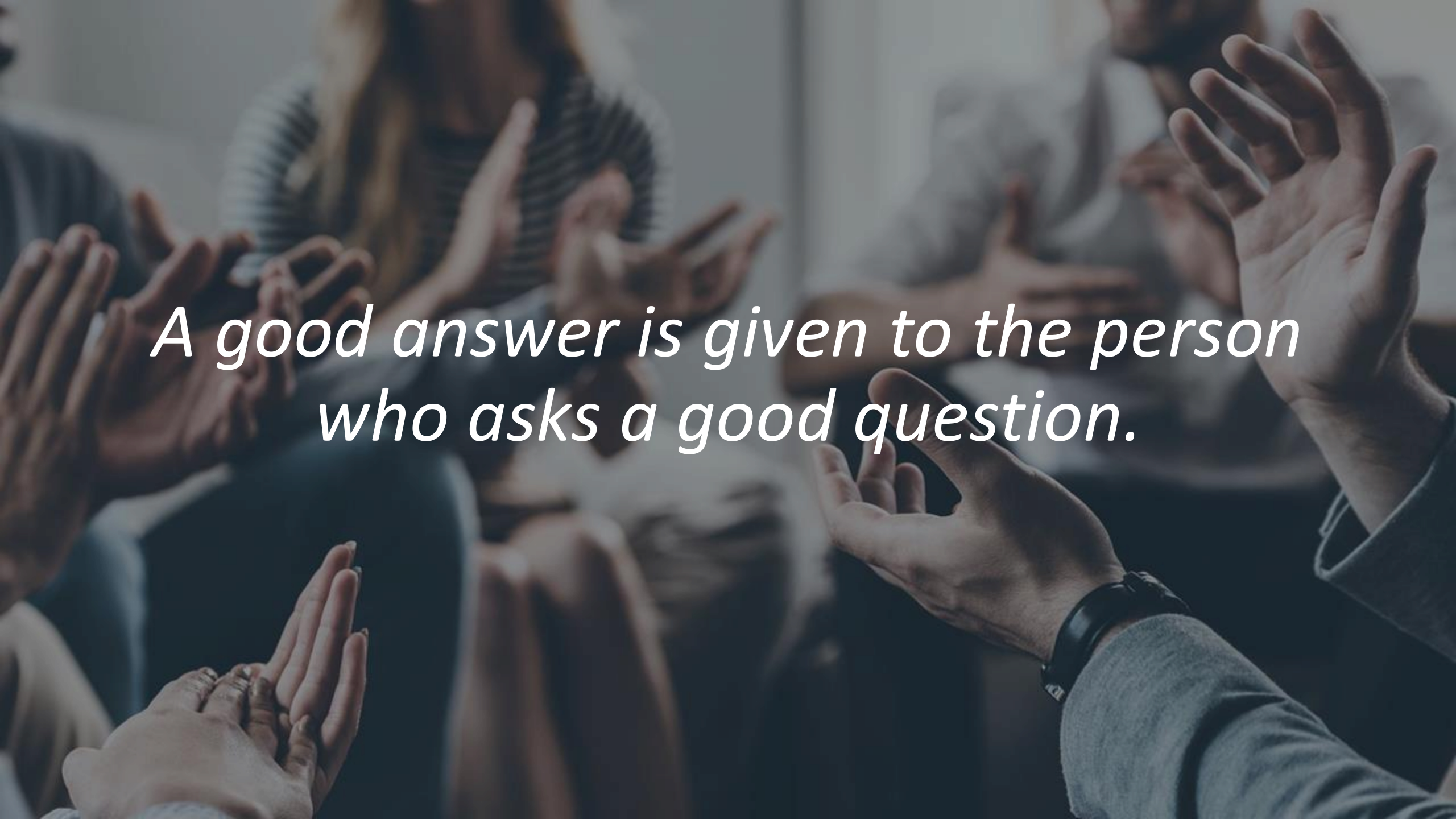


Explore applications

5



Prepare to create and refine prompts

A background image showing a group of people clapping their hands. The image is slightly blurred and has a dark, muted color palette. The hands are in various positions, some raised high, some lower, creating a sense of movement and celebration. The text is overlaid in the center of the image.

*A good answer is given to the person
who asks a good question.*

What is a Prompt?



Input/instruction given to an AI system



Analogy: Asking a professor vs searching Google



Examples:

Bad: 'Tell me about physics.'

Good: 'Explain Newton's 3 laws of motion with examples'



What is Prompt Engineering?

-  Designing prompts for better outputs
-  Like mastering Google search skills
-  Benefits:
 - Accuracy: precise answers
 - Productivity: faster insights
 - Creativity: content generation
 - Problem-solving: structured outputs

Why Prompt

Engineering Matters



 Law: Draft contracts, summarize cases



  Medicine: Summarize patient cases

 Business: Market analysis

 Education: Create quizzes, simplify notes



🎨 Arts: Poetry, storytelling



Key Terms



Prompt: Instruction/input



LLM: Large Language Model



Generative AI: Creates new content



Context: Background info shaping outputs

Examples Across Disciplines



Law: 'Summarize a legal case in 3 points'



Medicine: 'List diagnoses in plain terms'



Engineering: 'Explain how a circuit breaker works simply'



Business: 'Draft a renewable energy market analysis'



Literature: 'Write a Shakespearean sonnet about autumn'



Bad vs Good Prompt

✗ Bad Prompt:

'Tell me about physics.'

Output: Generic, vague response.

✓ Good Prompt:

'Explain Newton's laws with examples.'

Output: Structured, useful.

Live Demo

✨ Step 1: Vague prompt
→ output

✨ Step 2: Refined prompt
→ improved output

✨ Compare clarity and
depth



Recap



Prompt = input



Prompt engineering = better inputs for better outputs



Works across disciplines

Mini Quiz

- Q1: What is a prompt?
- Q2: Why is prompt engineering valuable?
- Q3: Which is better? 'Tell me about history' vs 'Summarize causes of WWI'

Google vs Generative AI



GOOGLE: RETRIEVES
DOCUMENTS

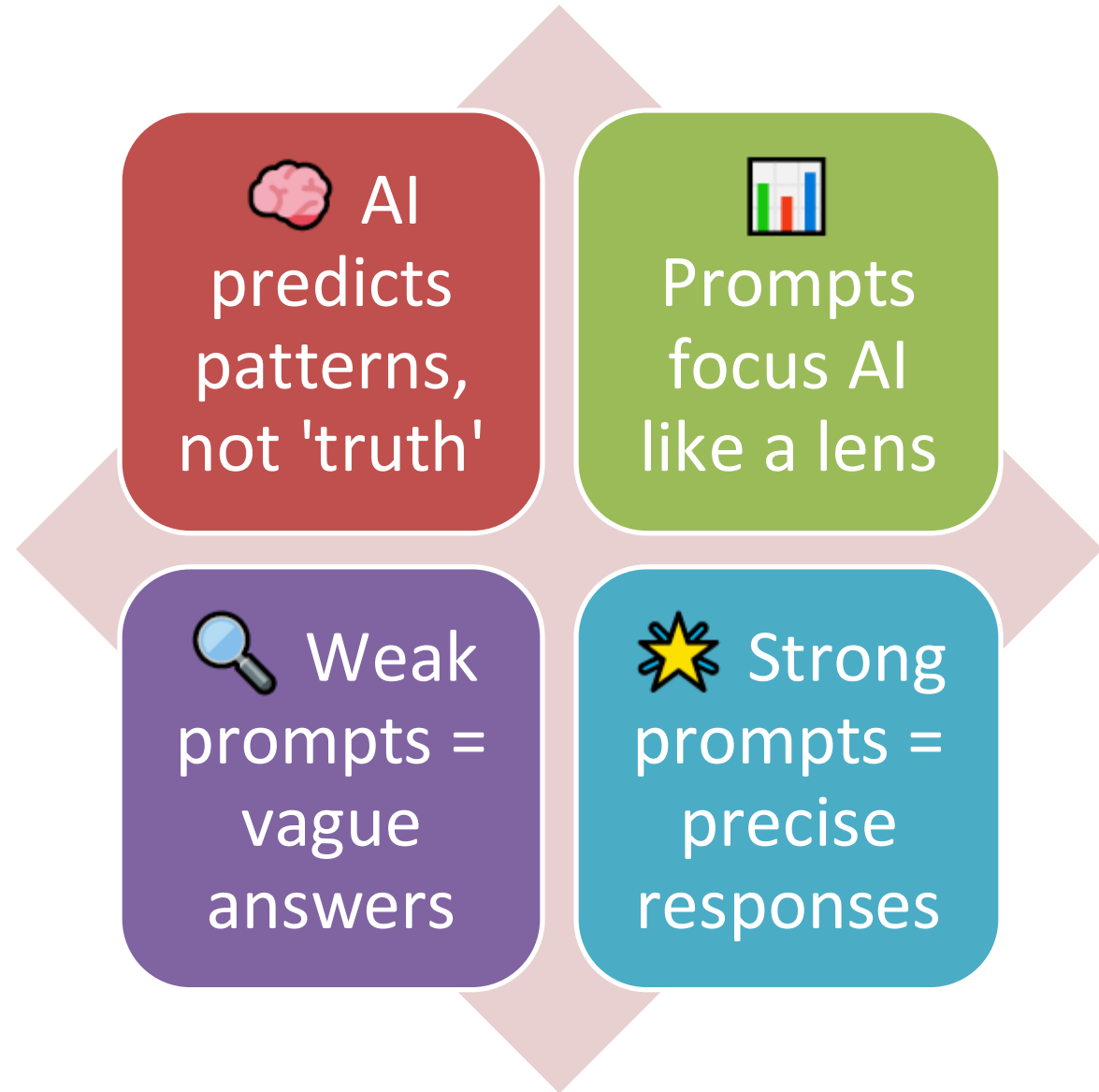


LLM: GENERATES
NEW CONTENT



RETRIEVAL VS
GENERATION

Why Prompts Shape Outputs



Prompt Impact Example

-  Broad: 'Summarize climate change'
-  Specific: 'Summarize the top 3 effects of climate change on agriculture in Turkey'

 Broad Prompt → Output: Very general, less useful.

 Specific Prompt → Output: Focused, practical insights.

Prompt Engineering is Iterative

-  Process: Prompt → Output → Refine → Output
-  Like editing drafts
-  Refinement improves clarity

Prompt

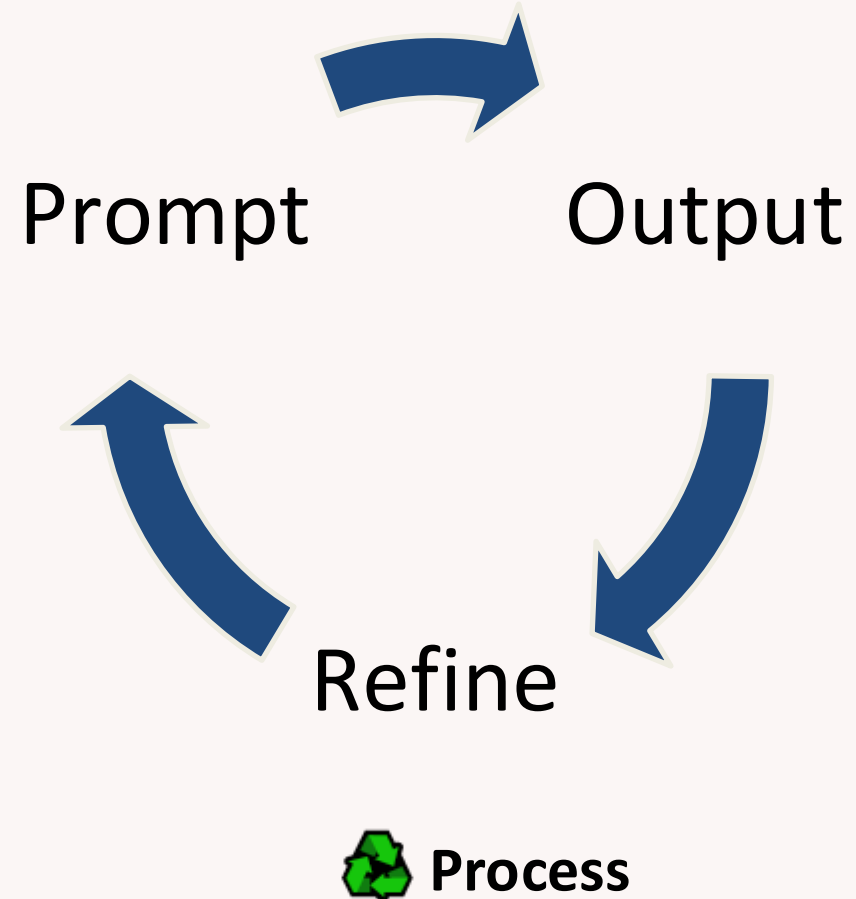
Output

Refined Prompt

Improved Output

Prompt Engineering is Iterative

- ✍️ Like editing drafts
- 💡 Refinement improves clarity



Everyday Uses



 **WRITING: ESSAYS,
REPORTS**



 **PRODUCTIVITY:
EMAILS, MEETING
NOTES**



 **LEARNING: SIMPLIFY
CONCEPTS, QUIZZES**



 **RESEARCH:
SUMMARIES,
HYPOTHESES**

Hands-On Demo

-  Ask a student for a task
-  Show vague prompt → output
-  Refine → improved output
-  Compare results

Discussion Prompt

💡 Where could YOU use prompt engineering in your field?

Wrap-Up



Prompt engineering
= future skill



No coding required



Improves
communication with AI

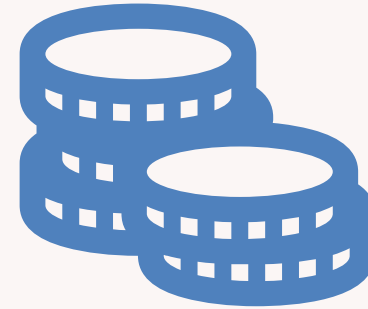


Next: How LLMs
work

Preview Next Week



 Week 2: How Large Language Models Work (Non-technical Overview)



 Topics: Tokens, embeddings, training, context windows